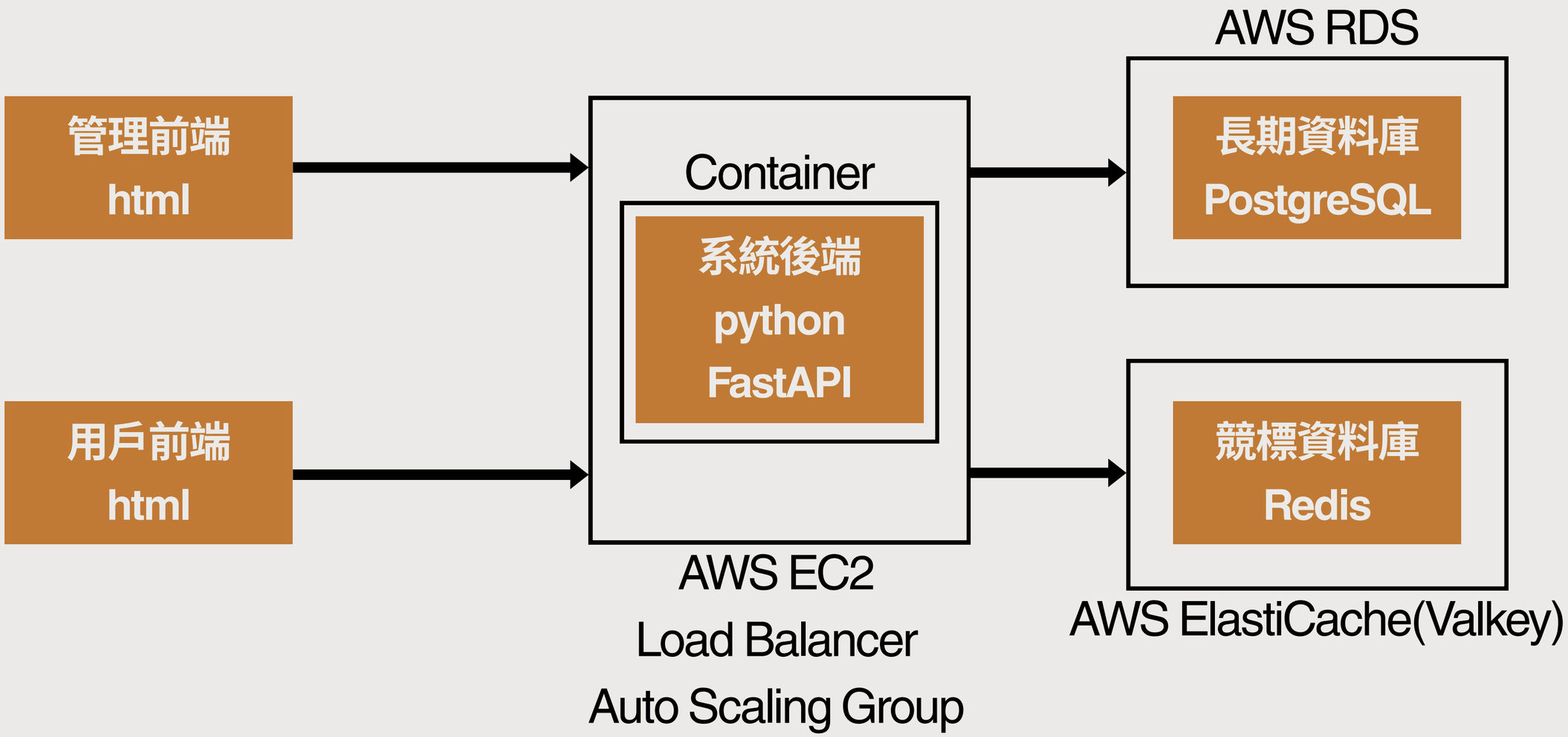
系統架構圖


管理前端
html
用戶前端
html
Container
系統後端
python
FastAPI
AWS EC2
Load Balancer
Auto Scaling Group
AWS RDS
長期資料庫
PostgreSQL
競標資料庫
Redis
AWS ElastiCache(Valkey)

核心技術說明

為了處理高併發寫入，在資料儲存上使用Redis，利用**記憶體**而非硬碟的機制縮短讀寫時間

- 前端傳入價格和用戶id
- 產品快取：管理者上架新產品時寫入Redis（取代舊產品）
- 出價：讀取Redis中的用戶權重和產品參數計算分數，再將出價寫入Redis覆蓋舊出價
- 排名：讀取Redis中的自動排序

Redis架構：

{bid:<product_id>}:ranking：儲存用戶id與當前分數，Redis ZSET提供排序功能（有TTL）

{bid:<product_id>}:details：儲存其餘的出價、時間（有TTL）

system:current_product：儲存現在產品資訊

user:{id}：儲存權重快取（競標結算時更新）

核心技術說明

產品結算時，將得標者寫入長期資料庫的winners表，每個得標者都有一筆紀錄

為了防止超賣，需要透過SQL的鎖來限制：第一個進行結算的人上鎖，之後的人只能讀取結算完成的資料（產品會被標記為已結算並同時更新Redis）

資料表架構支援上鎖：利用Bool結算欄位來確保（上鎖後）修改一次就無法改變

使用的平台、工具、套件

平台與工具

- Amazon Web Services
  - AWS EC2
  - AWS Auto Scaling Group (ASG)
  - AWS Application Load Balancer (ALB)
  - AWS ElastiCache: Redis
  - AWS Relational Database Services (RDS): PostgreSQL
  -
- Github
- Docker

使用的平台、工具、套件

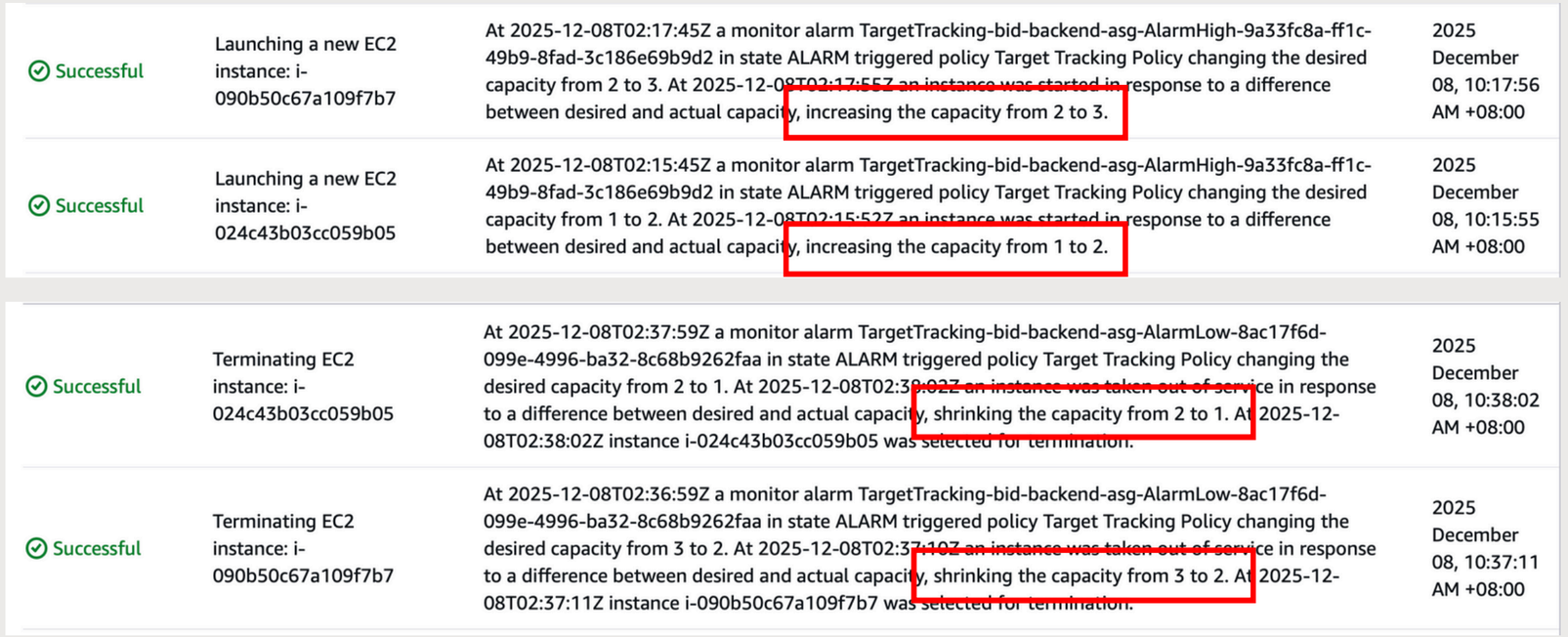
套件

- Backend
  - Python 3.13
  - FastAPI
  - Uvicorn: ASGI Server (for FastAPI)。
  - SQLAlchemy (for PostgreSQL)。
  - Redis Asyncio (for Redis)
  - Pydantic (for FastAPI)
- Testing
  - Locust
- Frontend
  - HTML

Scalability

Scalability 透過 AWS 的 Load Balancer 和 Auto Scaling Group 來達成，設定數據：

- 拓展觸發：CPU使用率平均 > 30%
- ASG：初始1台，上限3台



Successful	Launching a new EC2 instance: i-090b50c67a109f7b7	At 2025-12-08T02:17:45Z a monitor alarm TargetTracking-bid-backend-asg-AlarmHigh-9a33fc8a-ff1c-49b9-8fad-3c186e69b9d2 in state ALARM triggered policy Target Tracking Policy changing the desired capacity from 2 to 3. At 2025-12-08T02:17:55Z an instance was started in response to a difference between desired and actual capacity, increasing the capacity from 2 to 3.	2025 December 08, 10:17:56 AM +08:00
Successful	Launching a new EC2 instance: i-024c43b03cc059b05	At 2025-12-08T02:15:45Z a monitor alarm TargetTracking-bid-backend-asg-AlarmHigh-9a33fc8a-ff1c-49b9-8fad-3c186e69b9d2 in state ALARM triggered policy Target Tracking Policy changing the desired capacity from 1 to 2. At 2025-12-08T02:15:52Z an instance was started in response to a difference between desired and actual capacity, increasing the capacity from 1 to 2.	2025 December 08, 10:15:55 AM +08:00
Successful	Terminating EC2 instance: i-024c43b03cc059b05	At 2025-12-08T02:37:59Z a monitor alarm TargetTracking-bid-backend-asg-AlarmLow-8ac17f6d-099e-4996-ba32-8c68b9262faa in state ALARM triggered policy Target Tracking Policy changing the desired capacity from 2 to 1. At 2025-12-08T02:38:02Z an instance was taken out of service in response to a difference between desired and actual capacity, shrinking the capacity from 2 to 1. At 2025-12-08T02:38:02Z instance i-024c43b03cc059b05 was selected for termination.	2025 December 08, 10:38:02 AM +08:00
Successful	Terminating EC2 instance: i-090b50c67a109f7b7	At 2025-12-08T02:36:59Z a monitor alarm TargetTracking-bid-backend-asg-AlarmLow-8ac17f6d-099e-4996-ba32-8c68b9262faa in state ALARM triggered policy Target Tracking Policy changing the desired capacity from 3 to 2. At 2025-12-08T02:37:10Z an instance was taken out of service in response to a difference between desired and actual capacity, shrinking the capacity from 3 to 2. At 2025-12-08T02:37:11Z instance i-090b50c67a109f7b7 was selected for termination.	2025 December 08, 10:37:11 AM +08:00

測試設計

測試使用以下設定

- 2400名user註冊後競標
- 10分鐘，前4分鐘每秒增加10名user
- 指數增加，拓展完成後最初約 300-500 RPS ，最終階段提升為 700-900+ RPS

公式： Sleep = log(剩餘秒數 + 1) * 0.7 + 0.3

- log(600) ≈ 6.4 -> 4.7 秒 (剛開始)
- log(60) ≈ 4.1 -> 3.2 秒 (剩1分鐘)
- log(10) ≈ 2.3 -> 1.9 秒 (剩10秒)

測試數據（同影片）

在影片中，為了展示Auto Scaling，在每次bid時進行一次無意義運算來加重CPU使用率

巔峰期平均錯誤率（3 instances）：

從 600 RPS 增加到 900+ RPS 的100秒中，平均RPS = 689.11，Fail = 3，正確率 > **99.99%**

平均時間：(941.96*229178 - 1220.35*160267) / 68911 = **294.506 ms**

Type	Name	# Requests	# Fails	Median (ms)	95%ile (ms)	99%ile (ms)	Average (ms)	Min (ms)	Max (ms)	Average size (bytes)	Current RPS	Current Failures/s
POST	/api/bid	160267	13094	230	4600	7300	1220.35	194	10241	87.94	623.3	0

Type	Name	# Requests	# Fails	Median (ms)	95%ile (ms)	99%ile (ms)	Average (ms)	Min (ms)	Max (ms)	Average size (bytes)	Current RPS	Current Failures/s
POST	/api/bid	229178	13097	230	4500	5100	941.96	194	10241	87.03	914.1	0

影片